

LAB 15 RESULTS

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OVERVIEW

Objective

- Measure rise/fall times and minimum voltage during transition of a hex 555 timer and inverter circuit
- Compare measurements between good and bad circuit layouts
- Observe the significance of good and bad decoupling capacitor locations as well as return plane design

Measurement Differences

- Three identical circuits with good and bad practices are tested
- Good practices – Close decoupling capacitor, return plane
- Bad Decoupling – Far decoupling capacitor, return plane
- Bad practices – Far decoupling capacitor, no return plane

FINDINGS

With three identical base circuits but differing design elements, it is easy to observe the differences in measurements. This reveals the importance of good decoupling capacitors and a solid return plane that should be implemented in future designs.

	Rising Edge	
	Rise Time	Voltage Rail Collapse
Good Practices	1x	1x
Bad Decoupling	1.66x slower	2.19x slower
Bad Practices	2.06x slower	1.84x slower

	Falling Edge	
	Fall Time	Voltage Rail Collapse
Good Practices	1x	1x
Bad Decoupling	1.005x faster	4.34x slower
Bad Practices	1.94x slower	8.74x slower

The majority of the data supports the fact that good practices outperform bad practices. However, the voltage rail collapse for the rise edge of the bad practices ended up being better than the results of one bad decoupling capacitor. Measurements were repeated on multiple power supplies and oscilloscopes and revealed the same behavior. I believe this to be the result of a faulty section of the board. The scope 4 image illustrates the odd behavior.

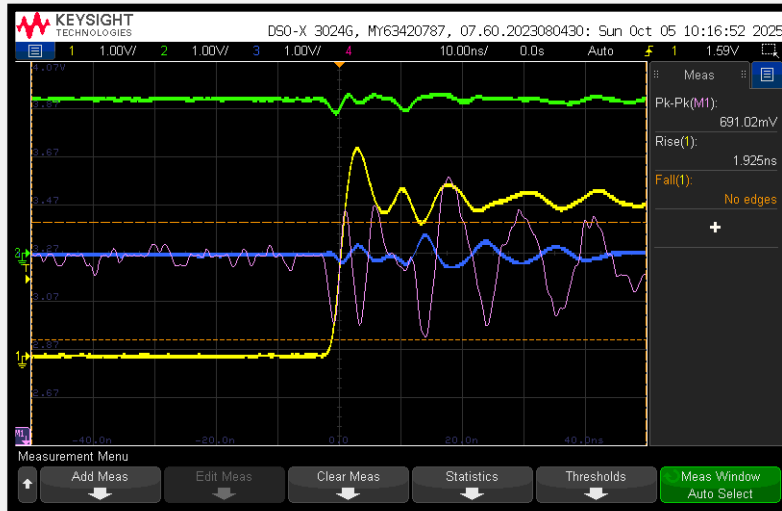
FULL MEASUREMENTS

Rising Edge:	Rise Time (ns)	Voltage Rail Collapse (V _{cc} – Return) (Pk-2-PkV)
Good Practices (Scope 0)	1.925 ns	0.69102V
Bad Decoupling Capacitor (Scope 2)	3.204 ns	1.5127V
Bad Practices (Scope 4)	3.973 ns	1.2715V

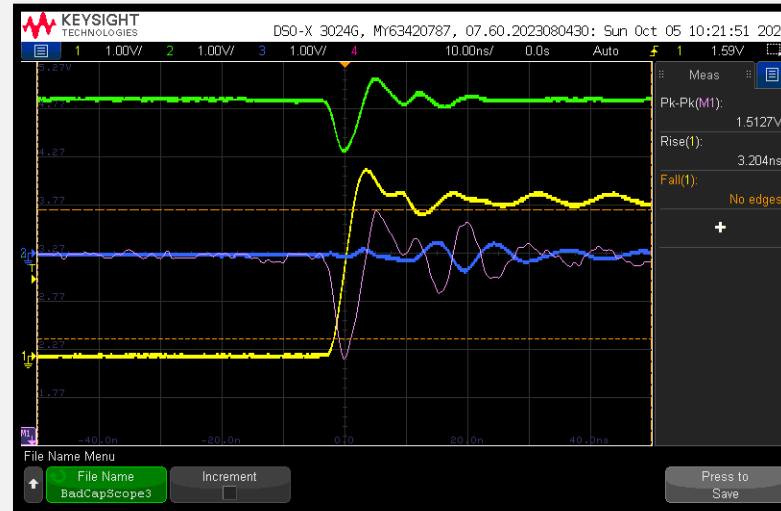
Falling Edge:	Fall Time (ns)	Voltage Rail Collapse (V _{cc} – Return) (Pk-2-PkV)
Good Practices (Scope 1)	2.240 ns	0.57500V
Bad Decoupling Capacitor (Scope 3)	2.228 ns	2.4951V
Bad Practices (Scope 4)	4.336 ns	5.0234V

Issues:	The Bad Practices quadrant of the board revealed odd behavior as observed in the scope images. I believe this affected measurements.
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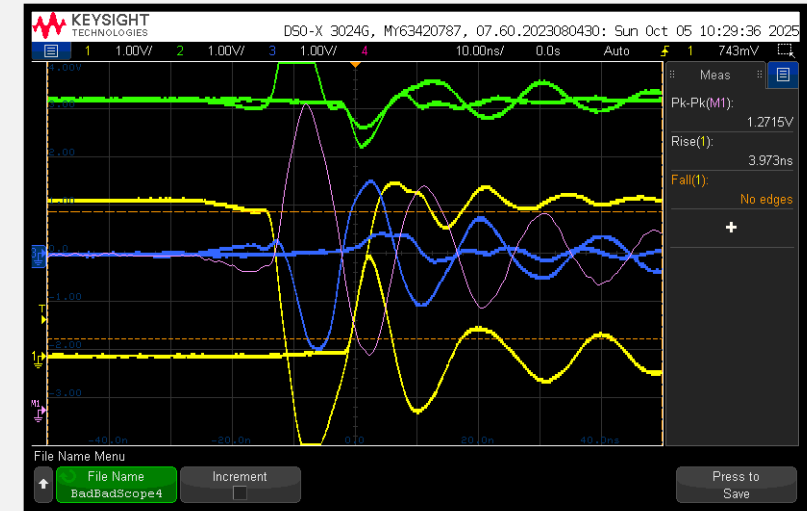
OSCILLOSCOPE IMAGES



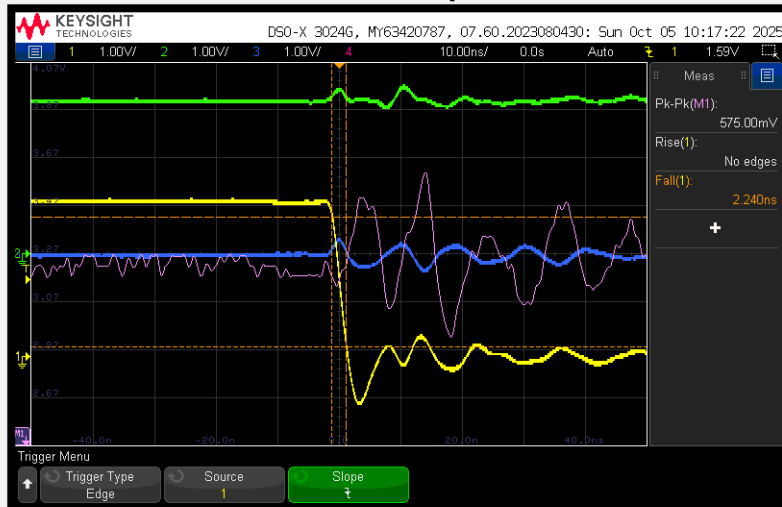
Good: Scope 0



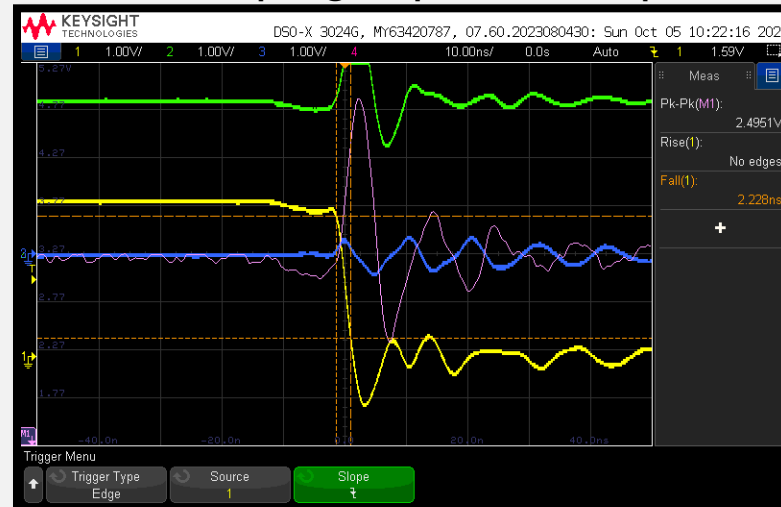
Bad Decoupling Capacitor: Scope 2



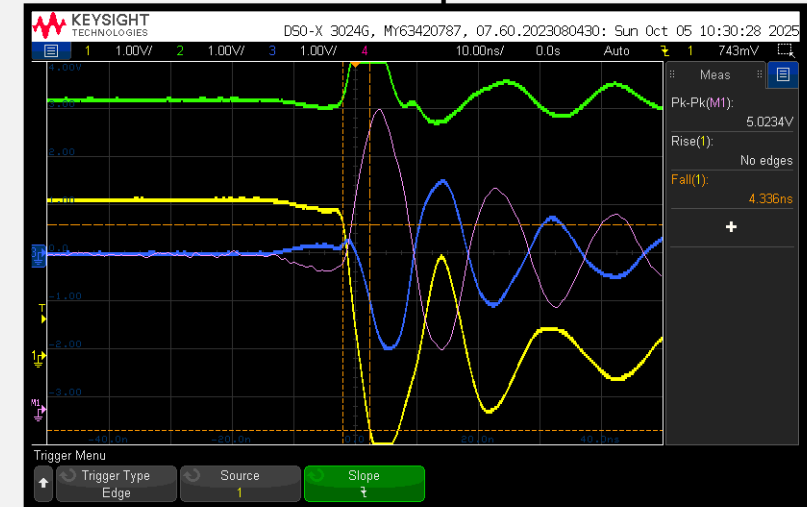
Bad: Scope 4



Good: Scope 1



Bad Decoupling Capacitor: Scope 3



Bad: Scope 5